CS5489 · 期中简答题与答案 / Written-Response Questions and Answers

37题：完整英文原题、中文题意、英文答案及中文对应答案。保留必要题图、作答图和成立条件，不收录详细解析。

简答包括判断说明、比较、计算推导、作图和综合问答；原题要求的理由、步骤与图均须作答。

沿用MT编号和组内顺序，目录与原卷索引使用本册页码。Gaussian Process题仍为历史拓展。

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MT011 · 识别过拟合，并提出可操作的改进

> **题型：** 简答题 / Written response · 解释、提出办法

考点： 学习流程与模型诊断

出处： 2020B期中Quiz Q11，原卷第5页（10分）

English question

What is overfitting? How do we know when it happens? Write down and describe several methods to address the problem of overfitting.

中文题意

什么是过拟合？怎样发现它？列举并说明几种解决过拟合的方法。

English answer

Overfitting occurs when a learned model captures training-specific noise or fluctuations that do not generalize. A much smaller training error than validation error is a warning sign, after checking that the data and preprocessing are comparable.

Use a simpler model to restrict the patterns it can fit; strengthen suitable regularization to discourage excessive weights; remove irrelevant features to reduce opportunities to fit noise; or collect more representative data so useful patterns recur across samples. Choose model complexity and regularization by cross-validation within the training data. Keep the final test set outside these choices.

中文对应答案

过拟合是指模型学到了训练数据中特有的噪声或偶然波动，这些规律无法推广到新数据。确认训练与验证数据的来源、预处理可比较后，如果训练误差远低于验证误差，就要警惕过拟合。

可以采用更简单的模型，限制它能拟合的模式；加强合适的正则化，抑制过大的权重；删除无关特征，减少拟合噪声的机会；或增加有代表性的数据，让真实规律在更多样本中重复出现。模型复杂度和正则强度应通过训练数据内部的交叉验证选择。最终测试集不参与这些选择。

MT020 · 数据很多，训练和测试仍都差

> **题型：** 简答题 / Written response · 诊断问题、提出办法

考点： 学习流程与模型诊断

出处： 2021A期中 Q7，原卷第3页（10分）

English question

Consider a situation where you have trained a linear classification model to diagnose mental disorders patients from symptom checklist data. You have 100,000 training examples, which is sufficient data for learning. However, your model performs poorly on both training and testing

datasets. What is the main problem that cause this situation? How to correct the problem? List 2 potential solutions.

中文题意

用症状清单训练线性分类器诊断精神障碍。已有100,000个训练样本，题设认为数据量充分，但训练和测试表现都很差。主要问题可能是什么？提出两种解决办法。

English answer

The intended diagnosis is underfitting: the representation or linear decision rule may be too restrictive. Two remedies are (1) add informative nonlinear or interaction features, and (2) use a suitable nonlinear classifier, such as a kernel SVM or a tree ensemble. If regularization is excessive, reducing it is another possibility. Verify optimization and label quality before attributing all poor fit to capacity.

中文对应答案

本题预期的判断是欠拟合：现有特征表示或线性决策规则可能过于受限。两种办法是：（1）加入有用的非线性特征或特征之间的交互项；（2）采用合适的非线性分类器，例如核SVM或树集成。若正则化过强，也可以减弱正则化。在把拟合差全部归因于模型表达能力之前，还要检查优化过程和标签质量。

MT022 · 五万客户、需要解释：选择哪种分类器?

> **题型：** 简答题 / Written response · 比较、提出办法

考点： 学习流程与模型诊断

出处： 2021A期中 Q9，原卷第3页（10分）

English question

Suppose you are a global market analyst who needs to analyse a product line (a group of similar products with different prices) and get some insight. You want to classify your 50,000 clients according to the product they brought. You have collected 25 features on your clients, e.g. age, income, educational level, frequency of using the product, their scores on other similar products. Among Logistic Regression, kernel SVM and random forest, which classifier will you prefer to use? List 2 reasons for using your selected classifier, and 1 reason for each of the other unselected classifiers.

中文题意

你是一名全球市场分析师，要分析一个产品系列（一组价格不同的相似产品），并获得业务洞见。你希望根据购买的产品，对50,000名客户进行分类。每位客户有25个特征，例如年龄、收入、教育程度、产品使用频率，以及对其他类似产品的评分。在逻辑回归、核SVM和随机森林中，你更倾向于选择哪一种？列出两个选择理由，并为另外两种模型各给出一个不选的理由。

English answer

A defensible choice, and the supplied answer, is **random forest**. It captures nonlinear relationships/interactions and supports feature-importance or permutation-importance analyses for inspection. A plain logistic model may miss nonlinear structure unless features are expanded. A conventional kernel SVM on 50,000 samples can incur substantial kernel computation/storage cost. These are task-based reasons, not proof that RF must be most accurate.

中文对应答案

一个合理的选择是**随机森林**，随卷答案也采用这一选择。它能够捕捉非线性关系和特征间的交互，并可通过特征重要性或置换重要性分析帮助解释模型。普通逻辑回归若不扩展特征，可能漏掉非线性结构；常规核SVM处理50,000个样本时，核计算与存储开销可能很大。这些理由说明随机森林适合该任务，但不能证明它的准确率一定最高。

MT046 · 完整讲出Gaussian NB

> **题型：** 简答题 / Written response · 解释

考点： Bayes、Naive Bayes与模型比较

出处： 2023A期中 Q7，原卷第4页（10分）；2025A期中 Q11，原卷第8页（10分）

English question

Consider the Naïve Bayes Gaussian classifier with feature vector $x \in \mathbb{R}^d$ and binary class label $y \in \{0, 1\}$.

(a) What are the assumptions of this classification model?

(b) What are the probability distributions in this model, and what form do they have?

(c) Given a feature vector x, what is the rule for performing classification with this model?

中文题意

考虑特征向量$x \in \mathbb{R}^d$、类别$y \in \{0, 1\}$的Gaussian NB分类器。(a) 模型有什么假设？ (b) 涉及哪些概率分布，各自是什么形式？ (c) 给定x，怎样分类?

English answer

(a) Training observations are modeled as independent and identically distributed. **Conditional on the class**, features are independent, and each feature has a Gaussian distribution. This does not require unconditional feature independence.

(b) Let $\pi = P(y = 1)$, with $P(y = 0) = 1 - \pi$. For class c,

$$p(x \mid y = c) = \prod_{j=1}^{d} \mathcal{N}(x_j; \mu_{cj}, \sigma_{cj}^2) = \mathcal{N}(x; \mu_c, \mathrm{diag}(\sigma_{c1}^2, \ldots, \sigma_{cd}^2)).$$

Here $\mu_c \in \mathbb{R}^d$ and each variance is positive. Unless a shared-variance model is specified, different classes may have different variances.

(c) Under equal misclassification costs, select the largest posterior:

$$\hat{y} = \arg\max_{c\in\{0,1\}} \left[\log P(y=c) + \sum_{j=1}^{d} \log \mathcal{N}(x_j; \mu_{cj}, \sigma_{cj}^2) \right].$$

The common evidence $p(x)$ cancels in this comparison. Use a fixed rule for ties.

中文对应答案

(a) 假设训练样本独立同分布；**给定类别后**，各特征相互独立，且每个特征服从高斯分布。这不要求特征在不区分类别时也独立。

(b) 记$\pi = P(y=1)$，则$P(y=0) = 1-\pi$。对类别c，有

$$p(x \mid y=c) = \prod_{j=1}^{d} \mathcal{N}(x_j; \mu_{cj}, \sigma_{cj}^2) = \mathcal{N}(x; \mu_c, \mathrm{diag}(\sigma_{c1}^2, \ldots, \sigma_{cd}^2)).$$

其中$\mu_c \in \mathbb{R}^d$，每个方差都大于0。除非题目指定共享方差，不同类别可以有不同的方差。

(c) 错分代价相同时，选择后验概率最大的类别：

$$\hat{y} = \arg\max_{c\in\{0,1\}} \left[\log P(y=c) + \sum_{j=1}^{d} \log \mathcal{N}(x_j; \mu_{cj}, \sigma_{cj}^2) \right].$$

比较时可以约去各类共同的分母$p(x)$；分数相同时，按预先固定的规则选择。

MT034 · 无限数据能消除分类错误吗

> **题型：** 简答题 / Written response · 判断并说明

考点： Bayes、Naive Bayes与模型比较

出处： 2021B*期中 Q8，原卷第3页（10分）

English question

Consider the following statement: If we train a Bayes classifier using infinite training data that satisfies all of the modeling assumptions (e.g., data is from the class-conditional densities, and sampled independently), then it will achieve zero training error over these training examples.

Is this statement true or false? Give your reasons.

中文题意

判断下述说法是否正确，并说明理由：如果用无限多、满足全部建模假设的训练数据来训练Bayes分类器（例如，数据来自相应的类条件分布，且各样本独立采集），那么它在这些训练样本上的错误率将为0。

English answer

False in general. Infinite correctly modeled data can remove parameter–estimation uncertainty, but it does not remove overlap between class distributions. At an input where both labels have positive posterior probability, even the optimal deterministic decision makes errors with conditional probability $1 - \max_c P(y = c \mid x)$.

Zero error requires additional separability conditions, such as disjoint class supports almost everywhere. Correct modeling alone does not guarantee this.

中文对应答案

一般不成立。 在模型设定正确时，无限数据可以消除参数估计的不确定性，却不能消除类别分布之间的重叠。如果某个输入对应的两种标签都有正的后验概率，即使采用最优的确定性决策，给定该输入时的错误概率仍为$1 - \max_c P(y = c \mid x)$。

要达到零错误，还需要额外的可分条件，例如各类别可能出现的输入范围，除概率为0的部分外互不重叠。仅有正确的模型设定不能保证零错误。

MT033 · 用图像特征解释两种建模方式

> **题型：** 简答题 / Written response · 比较、解释

考点： Bayes、Naive Bayes与模型比较

出处： 2021B*期中 Q7，原卷第3页（10分）

English question

Consider the classification problem of predicting a person's gender from an image of their face. Discuss the differences between adopting a discriminative classifier or a generative model for classification. How would you interpret the learned generative classifier and learned discriminative classifier?

中文题意

原题要求根据人脸图像预测题中记录的性别标签。比较采用判别式分类器与生成式模型的区别，并解释各自学到的模型表示什么。

English answer

A generative approach models the distribution of image features within each supplied class, $p(x \mid c)$, together with $p(c)$, and obtains a classifier by Bayes' rule. Its learned class-conditional model describes features typical of that class.

A discriminative approach learns $p(c \mid x)$ or a decision function directly. For a linear classifier, the weights describe which feature combinations help separate the supplied classes.

Discriminative training avoids specifying a full feature distribution. A simple generative model can be effective with limited labelled data when its assumptions are appropriate; this is a modelling trade-off, not a guarantee of better accuracy.

中文对应答案

生成式方法学习各个已给定类别中的图像特征分布$p(x \mid c)$，以及类别先验$p(c)$，再通过Bayes规则得到分类器。学到的类条件模型描述了该类常见的特征。

判别式方法直接学习$p(c \mid x)$或决策函数。在线性分类器中，权重反映哪些特征组合有助于区分这些类别。

判别式训练无需指定完整的特征分布；当假设合适时，简单的生成式模型在标注数据较少的情况下也可能有效。这是建模方式之间的取舍，不能据此保证哪一种准确率更高。

MT059 · LR与共享方差Gaussian NB

> **题型：** 简答题 / Written response · 比较

考点： Bayes、Naive Bayes与模型比较

出处： 2023B期中 Q8，原卷第5页（10分）

English question

Describe the major similarities and differences between logistic regression and naïve Bayes Gaussian classifier with shared variance parameter. List at least 5 similarities/differences (5 total).

中文题意

比较逻辑回归与共享方差参数的Gaussian NB，合计列出至少5项主要相同点或不同点。

English answer

1. Both can yield linear decision boundaries in the given feature space when Gaussian NB shares variances across classes.
2. Both can be learned using maximum likelihood, but LR maximizes conditional label likelihood, whereas NB fits the joint model through class priors and class-conditional likelihoods.
3. NB is generative; LR directly models the class posterior and is discriminative.
4. Both have multiclass extensions and can incorporate regularization or parameter priors.

5. For binary classification with d features, LR has $d+1$ coefficients, including the intercept. Under the key's **single shared scalar variance** convention, NB has $2d+1$ class–conditional parameters: two mean vectors and one variance. A learned binary class prior adds one more free parameter, making $2d+2$ in total.

If "shared variance" means one shared variance **per feature**, NB has $3d+1$ free parameters including the learned prior. State which model and counting convention you use.

中文对应答案

1. 当Gaussian NB在类别之间共享方差时，两种模型都能在给定特征空间中产生线性决策边界。
2. 两者都可以用最大似然估计训练，但逻辑回归最大化标签的条件似然；NB则通过类别先验和类条件似然拟合联合模型。
3. NB是生成式模型；逻辑回归直接建模类别后验，是判别式模型。
4. 两者都能扩展到多分类，也都可以加入正则化或参数先验。
5. 对具有d个特征的二分类任务，逻辑回归有$d+1$个系数，其中包括截距。按原解采用的**单个共享标量方差**约定，NB有$2d+1$个类条件参数，即两个均值向量和一个方差。若还学习二分类的类别先验，就再增加一个自由参数，总数为$2d+2$。

如果"共享方差"指的是**每个特征各有一个类间共享的方差**，那么NB连同可学习先验共有$3d+1$个自由参数。作答时应说明采用哪种模型和计数约定。

MT021 · 同为线性分类器，学习原则有何不同

> **题型：** 简答题 / Written response · 比较

考点： Bayes、Naive Bayes与模型比较

出处： 2021A期中 Q8，原卷第3页（10分）

English question

We have seen several types of linear classifiers, such as Logistic Regression, SVM, and Naïve Bayes with Gaussian CCDs with the same variance, which can all be expressed as learning the same function $f(x) = w^T x + b$. How are these linear classifiers different? Which linear classifier is best in terms of accuracy?

中文题意

逻辑回归、SVM、各类共享方差的Gaussian NB都能写成学习$f(x) = w^T x + b$。它们的区别是什么？哪一种的准确率最好?

English answer

Their decision functions can share a linear form while their training objectives differ. Logistic regression maximizes the conditional likelihood of the observed labels, usually with regularization. A standard SVM trades off margin size against hinge-loss penalties. Gaussian NB fits class-conditional distributions and class priors, then obtains a linear decision rule under shared class variances.

There is no universally most accurate choice. Performance depends on the data, whether model assumptions fit, sample size and regularization. Compare suitable candidates using held-out validation or cross-validation, and evaluate the selected procedure on an untouched test set.

中文对应答案

这些模型的决策函数可以具有相同的线性形式，但训练目标不同。逻辑回归最大化已观测标签的条件似然，通常还会加入正则化；标准SVM在间隔大小与hinge损失惩罚之间作权衡；Gaussian NB拟合类条件分布和类别先验，在类间共享方差时得到线性决策规则。

没有哪种模型在所有任务上都最准确。表现取决于数据、模型假设是否合适、样本量和正则化。应通过留出的验证集或交叉验证比较候选方案，再用未参与选择的测试集评价最终方案。

MT058 · 智能笔上的文本NB如何改进

> **题型：** 简答题 / Written response · 提出办法

考点： Bayes、Naive Bayes与模型比较

出处： 2023B期中 Q7，原卷第4页（10分）

English question

You are working for a “smart pen” company, which can scan text using a handheld pen device. Your job is to build a model to classify the text the pen reads into different sentiment categories, such as “happy”, “sad”, “neutral”, which will then turn on different lights on the pen as a special effect. Because of the limited processing power on the pen, you will use a generative model classifier with bag-of-words model. A training set is provided, consisting of 5000 sentences and their class labels. Your initial classifier is Naïve Bayes Bernoulli model, but the performance is poor. How could you improve the model performance? (List 4 improvements)

中文题意

智能笔扫描文字后，要把它分成happy、sad、neutral等情绪类别，并点亮不同的灯。设备算力有限，采用词袋和生成式分类器。已有5000句带类别标签的训练文本，初始Bernoulli NB表现较差。提出4项改进。

English answer

1. Try Multinomial NB with word counts.
2. Add selected bigrams, such as “not happy”.
3. Tune additive smoothing on validation data.
4. Select a compact, informative vocabulary.

Validate accuracy and device memory/inference cost.

中文对应答案

1. 尝试用词频作为输入的多项式朴素贝叶斯（Multinomial NB）。
2. 加入筛选过的相邻双词组合，例如“not happy”。
3. 用验证数据调整加法平滑的强度。
4. 选择规模较小、信息量足的词表。

同时验证准确率，以及设备上的内存占用和推理开销。

MT047 · 解释Logistic目标中的两项

> **题型：** 简答题 / Written response · 解释

考点： 逻辑回归与参数正则

出处： 2023A期中 Q8，原卷第5页（10分）；2025A期中 Q8，原卷第5页（10分）

English question

For binary classification, the logistic regression model is trained using the following objective function:

$$(w^*, b^*) = \arg\min_{w,b} \left\{ \alpha w^T w + \sum_{i=1}^{N} \log(1 + \exp[-y_i f(x_i)]) \right\}.$$

(a) Explain the purpose of each term in the objective function.

(b) Discuss the meaning of the hyperparameter α in logistic regression. How does varying α influence the values of weights w and the performance of the model?

中文题意

二分类逻辑回归的训练目标为

$$(w^*, b^*) = \arg\min_{w,b} \left\{ \alpha w^T w + \sum_{i=1}^{N} \log(1 + \exp[-y_i f(x_i)]) \right\}.$$

(a) 分别解释每项的作用。(b) 解释α的意义及其变化对w和模型表现的影响。

English answer

The first term is L2 regularization; it discourages large weights. The sum is the logistic negative conditional log-likelihood for labels $y_i \in \{-1,+1\}$; it rewards assigning higher probability to the true label. Larger α strengthens shrinkage and may underfit; smaller α permits larger weights and may overfit. Select α by validation. Under a zero-mean isotropic Gaussian weight prior, α is proportional to the inverse prior variance.

中文对应答案

第一项是L2正则项，用来抑制大权重。求和项是标签$y_i \in \{-1,+1\}$时，逻辑回归的负条件对数似然；它鼓励模型给真实标签更高的概率。α越大，权重收缩越强，可能欠拟合；α越小，允许的权重越大，可能过拟合。应通过验证选择α。若权重先验为零均值、各向同性的高斯分布，α与先验方差的倒数成正比。

MT024 · 损失与正则项：α从哪里来?

> **题型：** 简答题 / Written response · 解释

考点： 逻辑回归与参数正则

出处： 2021A期中 Q11，原卷第4页（10分）

English question

The goal of supervised training is to find the weights that best fit the training data (X,Y). To determine what we mean by best fit, we introduce a loss function L and the goal is to find the weights that minimize:

$$\hat{w} = \arg\min_w \{L(X,Y) + \alpha\|w\|^2\}.$$

(a) What is the purpose of the term $\|w\|^2$?

(b) How is α determined?

(c) Suppose we replace $\|w\|^2$ with $\|w\|_1$. What will be the effect when training the classifier?

中文题意

监督训练的目标是找到最能拟合训练数据(X,Y)的权重。用损失函数L衡量拟合程度，训练目标为

$$\hat{w} = \arg\min_w \{L(X,Y) + \alpha\|w\|^2\}.$$

(a) $\|w\|^2$这一项的目的是什么？(b) 怎样确定α？(c) 将$\|w\|^2$换成$\|w\|_1$，对分类器的训练会有什么影响?

English answer

(a) The quadratic term discourages large weights and controls complexity. (b) Treat α as a hyperparameter and select it by validation or cross-validation inside the training data. (c) L1 encourages sparse coefficients and can perform feature selection; exact zeros depend on the data and penalty strength. Keep preprocessing inside each training fold and refit after selection.

中文对应答案

(a) 二次项抑制大权重，从而控制模型复杂度。

(b) 将α作为超参数，在训练数据内部通过验证或交叉验证选择。

(c) L1鼓励稀疏系数，因此可以用于特征选择；系数是否恰好为0，取决于数据和惩罚强度。预处理也应在各个训练折内部完成；选好设置后，再重新拟合最终模型。

MT012 · L1逻辑回归：写目标并画边界

> **题型：** 简答题 / Written response · 列式、作图、解释

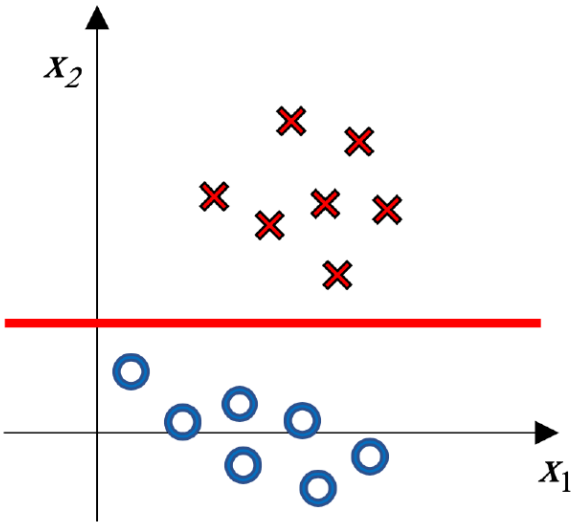
考点： 逻辑回归与参数正则

出处： 2020B期中Quiz Q12，原卷第6页（10分）

English question

(a) Consider a logistic regression classifier using L1 regularization instead of L2 regularization? Write down the optimization problem for learning the classifier.

(b) Suppose we have the data in the figure below. Draw the most likely decision boundary for the L1-regularized logistic regression classifier. You can move/reshape the red line provided below. Explain why this should be the decision boundary.



原卷提供一条可调整红线；按题意自行移动或重画。 / The original supplies an adjustable red line.

中文题意

(a) 将逻辑回归的L2正则换成L1，写出训练优化问题。

(b) 观察题图，画出最可能的L1正则逻辑回归边界，并解释。可以移动或调整图中给出的红线。

English answer

(a) For labels $y_i \in \{-1, +1\}$, one consistent objective is

$$(w^*, b^*) = \arg\min_{w,b} \left\{ \frac{1}{C} \sum_j |w_j| + \sum_i \log\left(1 + e^{-y_i(w^T x_i + b)}\right) \right\}, \quad C > 0.$$

(b) Draw a horizontal line in the gap between the upper red crosses and lower blue circles, as the supplied red line does. The classes mainly differ in x_2, so a natural sparse solution keeps x_2 and sets $w_1 = 0$. Its boundary is $x_2 = -b/w_2$ with $w_2 \neq 0$; the figure does not determine a unique height. L1 encourages exact zeros, but does not guarantee an axis-parallel boundary for every dataset or C.

中文对应答案

(a) 对标签$y_i \in \{-1, +1\}$，一种一致的目标函数写法是

$$(w^*, b^*) = \arg\min_{w,b} \left\{ \frac{1}{C} \sum_j |w_j| + \sum_i \log\left(1 + e^{-y_i(w^T x_i + b)}\right) \right\}, \quad C > 0.$$

(b) 在上方红叉与下方蓝圈之间的空隙画水平线，题图给出的红线就是这种画法。两类主要沿x_2方向分开，因此一个自然的稀疏解是保留x_2、令$w_1 = 0$。边界为$x_2 = -b/w_2$，其中$w_2 \neq 0$；原图无法确定唯一高度。L1鼓励系数恰好为0，但不能保证在每个数据集或每个C下，边界都与坐标轴平行。

MT048 · 用一个例子讲清Kernel Trick

> **题型：** 简答题 / Written response · 解释、举例

考点： SVM与核方法

出处： 2023A期中 Q9，原卷第6页（10分）

English question

Explain the concept of the "kernel trick" in Support Vector Machines (SVMs) and provide an intuitive example (e.g., using a figure).

中文题意

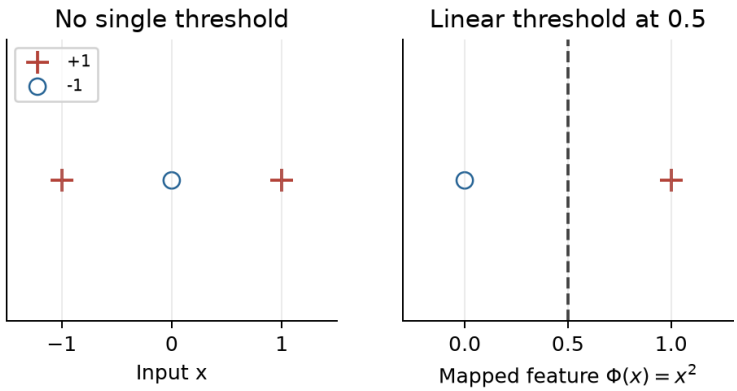
解释SVM中的kernel trick，并给一个直观例子，可以画图。

English answer

A feature map can make data linearly separable. The kernel trick replaces mapped inner products by k(x,z), without constructing the mapped vectors. For example, x=−1,+1 are positive and x=0 negative. Mapping Φ(x)=x^2 separates them at 0.5; its kernel is k(x,z)=x^2z^2.

中文对应答案

特征映射可以让数据变得线性可分。核技巧用$k(x,z)$代替映射后向量的内积，无需实际构造这些向量。例如，$x=-1,+1$属于正类，$x=0$属于负类。映射$\Phi(x)=x^2$后，可以在0.5处设阈值分开两类；对应的核为$k(x,z)=x^2z^2$。



答案示例图 / Answer example

MT060 · 二次核SVM：大C与小C的边界

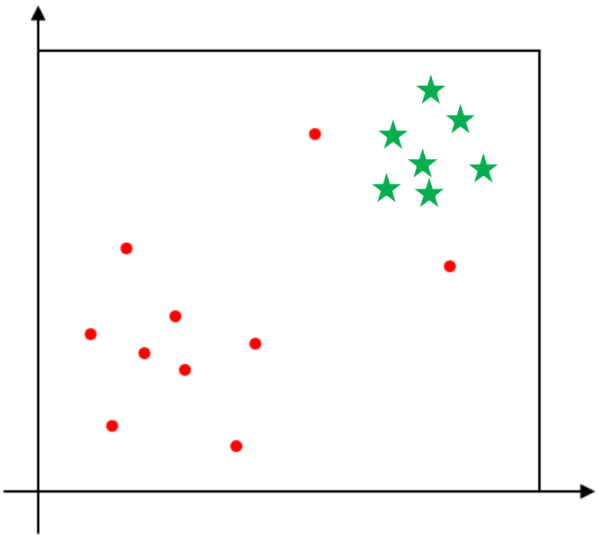
> **题型：** 简答题 / Written response · 作图、比较、解释

考点： SVM与核方法

出处： 2023B期中 Q9，原卷第6页（10分）

English question

Consider the training data in the below figure, and the kernel SVM with quadratic polynomial kernel function. (a) Draw the decision boundary when C → ∞? (b) Draw the decision boundary for C → 0? (c) Which decision boundary would be better on the test data? For each part, give your reasons.



中文题意

给定题图训练点，使用二次多项式核SVM。(a) C趋于无穷时画边界。(b) C趋于0时画边界。(c) 哪个在测试数据上可能更好？每问说明理由。

English answer

(a) A **large** C strongly penalizes margin violations. The supplied left–hand sketch bends the boundary to accommodate the isolated red points. Zero training error in this limit requires separability in the chosen quadratic feature space.

(b) For the **strict limit** $C \to 0^+$, use the standard equally weighted soft–margin objective with an unpenalized bias. The figure has ten red and seven green points. Encode red as -1 and green as $+1$. The candidate $w = 0, b = -1$ has objective $14C$, so the optimum satisfies $\|w_C\|^2 \le 28C$ and $w_C \to 0$. Minimizing the remaining hinge loss selects $b_C \to -1$, the majority–class sign. Thus the plotted region is eventually classified entirely red, with no separating boundary inside it. The supplied right–hand curve illustrates a smaller positive C, not this strict limit.

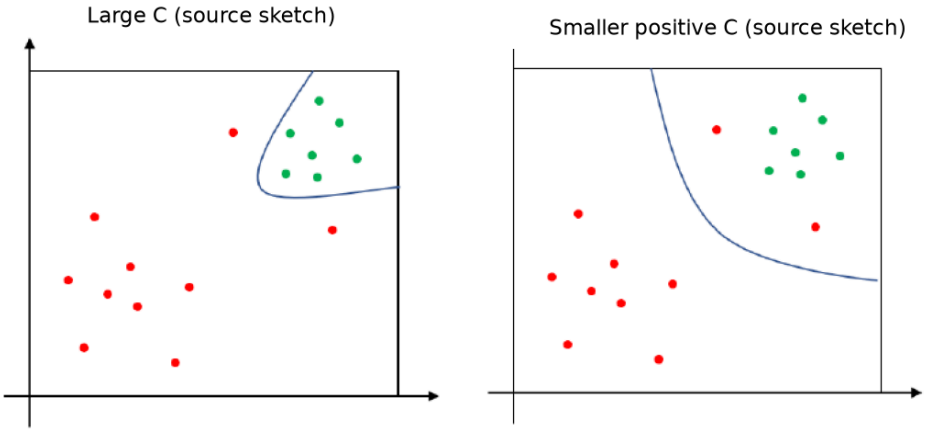
(c) If the isolated points are noise, a moderate positive C may generalize better than fitting them closely. This does not mean the all–red limit is best. A training diagram alone cannot determine test accuracy; choose C using validation data.

中文对应答案

(a) C**较大**时，违反间隔约束的代价很高。随卷答案左图的边界会弯曲，以适应孤立的红点。这个极限能达到零训练误差，前提是数据在所选二次特征空间中可分。

(b) 对**严格的**$C \to 0^+$**极限**，采用各样本权重相同、偏置不受正则惩罚的标准软间隔目标。原图有10个红点、7个绿点。令红类为-1、绿类为$+1$，候选解$w = 0, b = -1$的目标值为$14C$，因此最优解满足$\|w_C\|^2 \le 28C$，权重趋于0。继续使hinge损失最小，会让$b_C \to -1$，即多数类的符号。因此，图示区域最终全部判为红类，区域内不再有分隔边界。随卷右图表示较小的正C，不是这个严格极限。

(c) 如果这些孤立点是噪声，取适中的正C可能比努力拟合它们更容易推广到新数据，但不能因此说全判红类的极限最好。仅凭训练图无法确定测试准确率，应在验证数据上选择C。



随卷答案草图：右图表示较小正C，不表示严格$C \to 0^+$。 / Supplied sketches: the right plot represents a smaller positive C, not the strict $C \to 0^+$ limit.

原题差异 / Source note： 原解把较小正C的分隔曲线用于C趋零；按本题标准目标，严格极限全判多数红类。 / The supplied small-positive-C sketch is not the strict limit, which predicts the majority red class under the stated standard objective.

MT010 · 同一张分布图，Gaussian Bayes与二次核SVM如何判断?

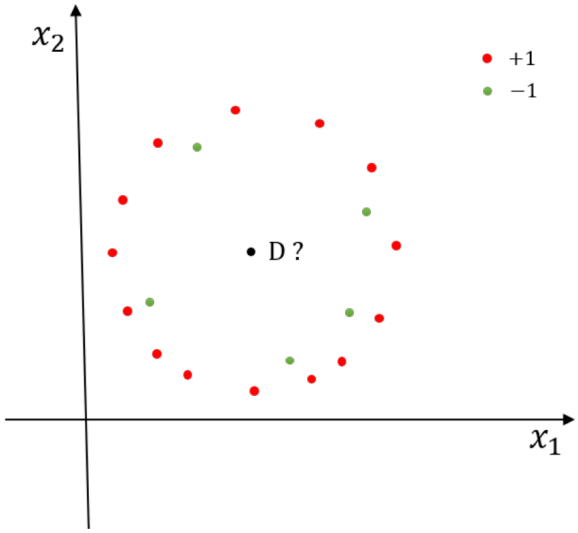
> **题型：** 简答题 / Written response · 读图或公式、比较

考点： SVM与核方法

出处： 2020B期中Quiz Q10，原卷第5页（10分）

English question

Consider the data distribution shown in the figure below. If you train a Bayes Gaussian classifier, which class will be predicted for data point D? If we change the classifier to an SVM with polynomial kernel of degree 2, which class will be predicted? Explain why.



中文题意

依据题图的两类训练数据，Gaussian Bayes分类器会将点D判为哪一类？换成二次多项式核SVM后呢？分别解释。红点为+1，绿点为−1。

English answer

The supplied solution expects **+1 for Gaussian Bayes and −1 for quadratic-kernel SVM**. Bayes compares $p(D \mid c)P(c)$. The more numerous red points give a larger estimated red-class prior, which can offset a density disadvantage at D; the supplied schematic fit favours red. It is not simply a nearest-centre decision. The quadratic SVM can enclose the inner green points with a curved boundary, placing D on their negative side.

These are the intended schematic results: exact Gaussian parameters and SVM C are not supplied, so the plot does not uniquely determine a numerical fit.

中文对应答案

随卷解答预期：**Gaussian Bayes判为+1，二次核SVM判为−1。** Bayes比较$p(D \mid c)P(c)$。红点更多，会使估计的红类先验更大，可能抵消D处类条件密度较低的影响；随卷示意拟合最终倾向红类。它不是简单地选择最近的类别中心。二次核SVM可以用曲线包围内部绿点，把D也放在绿色负类一侧。

这些是示意图所表达的判断。题目没有给出确切的高斯参数和SVM的C，因此不能从图中唯一确定一个数值拟合结果。

MT009 · 十万维特征、两千样本：SVM怎样提速?

> **题型：** 简答题 / Written response · 提出办法、解释

考点： SVM与核方法

出处： 2020B期中Quiz Q9，原卷第4页（10分）

English question

Your friend is training a linear SVM on a patient data from a hospital. There are 2000 patients, and each patient has 100,000 features extracted from their time in the hospital. Your friend complains that training the linear SVM takes too long. What suggestions do you give your friend to speed up the training?

中文题意

医院数据有2000名患者，每人100,000维特征。朋友训练线性SVM很慢，你建议怎样提速？说明理由。

English answer

Compare primal and dual formulations and use a solver suited to the data. The primal has about 100,000 weights plus a bias; the dual has 2,000 sample multipliers and a 2,000×2,000 Gram matrix. A dual method may therefore be attractive when d≫N. Also exploit sparse inputs and efficient linear-SVM implementations; validated feature reduction is another option. Variable counts alone do not guarantee a wall-clock speedup.

中文对应答案

比较原始形式和对偶形式，选择适合数据的求解器。原始形式约有100,000个权重和一个偏置；对偶形式有2,000个样本乘子，以及一个2000×2000的Gram矩阵。因此，当$d \gg N$时，对偶方法可能更合适。还可利用输入的稀疏性和高效的线性SVM实现，或尝试经验证有效的特征降维。仅比较变量数量，不能保证实际运行时间一定缩短。

MT061 · 预测内存：线性SVM与RBF SVM

> **题型：** 简答题 / Written response · 排序、解释

考点： SVM与核方法

出处： 2023B期中 Q10，原卷第7页（10分）

English question

Consider a binary classification problem where the input feature space is d=1,000 dimensions, and we have 2,000 training samples. Rank the following classifiers based on their memory usage (from lowest to highest) when implementing their respective prediction functions: linear SVM with C=0.1, linear SVM with C=1000, RBF kernel SVM with C=0.1, RBF kernel SVM with C=1000. Explain the reasons for your rankings.

中文题意

二分类输入1000维、训练样本2000条。按预测函数的内存从小到大排列：线性SVM(C=0.1)、线性SVM(C=1000)、RBF SVM(C=0.1)、RBF SVM(C=1000)。解释排序。

English answer

The two linear models tie under the standard compressed representation: d weights plus one bias, **1001 numbers**, independent of C. A kernel model with M support vectors stores approximately M(d+1)+1 numbers (support–vector coordinates, one coefficient each, and a bias).

Under this representation, the ranking is linear(C=0.1) = linear(C=1000) < ordinary nondegenerate RBF models. The two RBF models cannot be universally ordered from C alone; their actual support–vector counts are needed. The supplied answer puts C=1000 before C=0.1 based on a presumed decrease in M, but support–vector count is not generally monotone in C.

中文对应答案

按标准的压缩存储方式，两个线性模型并列：都只需保存d个权重和一个偏置，共**1001个数**，与C无关。具有M个支持向量的核模型约需保存$M(d+1)+1$个数，包括支持向量坐标、每个支持向量的一个系数，以及一个偏置。

在这种表示下，内存占用排序为：线性（C=0.1）=线性（C=1000）<通常的非退化RBF模型。两个RBF模型不能仅凭C确定先后，还要知道实际支持向量数量。原解把C=1000排在C=0.1之前，依据是支持向量数量M会减少的假设；但M一般不随C单调变化。

MT025 · 绝对残差损失对异常值有什么影响?

> **题型：** 简答题 / Written response · 解释

考点： 回归、特征选择与残差损失

出处： 2021A期中 Q12，原卷第4页（10分）

English question

Consider a linear regression function $f(x) = w^T x + b$, which is trained on the data (x_i, y_i), i=1,…,N, using the following optimization problem, where the absolute prediction error is the loss function:

$$(\hat{w}, \hat{b}) = \arg\min_{w,b} \sum_{i=1}^{N} |f(x_i) - y_i|.$$

What will be the effect of using this loss during training? Why?

中文题意

考虑线性回归函数$f(x) = w^T x + b$，在数据(x_i, y_i)、$i = 1, \ldots, N$上训练，以预测误差的绝对值作为损失：

$$(\hat{w}, \hat{b}) = \arg\min_{w,b} \sum_{i=1}^{N} |f(x_i) - y_i|.$$

使用这种损失会怎样影响训练？为什么？

English answer

Absolute–error loss grows linearly instead of quadratically, so very large output residuals have less relative influence than under least squares. Its residual derivative has constant magnitude away from zero, making it more robust to large y–outliers. This is a change to the data loss, not an L1 penalty on the weights, and it does not imply sparse coefficients.

中文对应答案

绝对误差损失随残差线性增长，而不是平方增长，因此相对于最小二乘，特别大的输出残差影响较小。除残差为0的位置外，它对残差的导数绝对值恒定，所以对输出y中的大异常值更稳健。这里改变的是数据损失，并不是给权重加L1惩罚，也不意味着会得到稀疏系数。

MT063 · Huber损失：画图并解释好处

> **题型：** 简答题 / Written response · 作图、解释

考点： 回归、特征选择与残差损失

出处： 2023B期中 Q12，原卷第9页（10分）

English question

Consider linear regression using the Huber loss function:

$$L(f(x_i), y_i) = \begin{cases} \frac{1}{2}[y_i - f(x_i)]^2, & |y_i - f(x_i)| < 1, \\ |y_i - f(x_i)| - \frac{1}{2}, & \text{otherwise.} \end{cases}$$

(a) Draw a plot of the loss function.

(b) Explain one benefit of using Huber loss for linear regression.

中文题意

令残差 $r = y_i - f(x_i)$，线性回归使用

$$L(r) = \begin{cases} \frac{1}{2}r^2, & |r| < 1, \\ |r| - \frac{1}{2}, & \text{otherwise.} \end{cases}$$

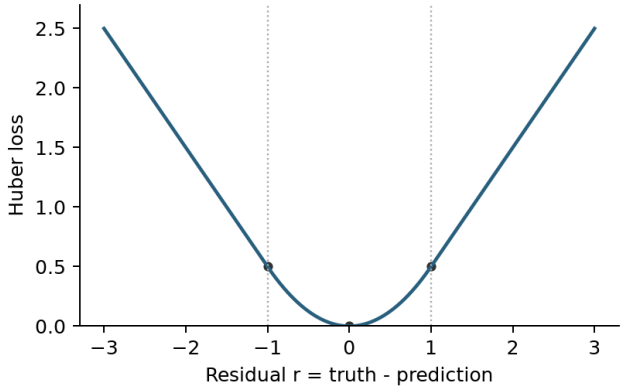
(a) 画出损失函数。(b) 解释使用Huber损失的一个好处。

English answer

Draw a parabola between −1 and 1, joined smoothly to straight tails: L(0)=0, L(±1)=1/2 and L(±2)=3/2. Its residual derivative is r inside the interval and sign(r) outside. Thus extreme output residuals have bounded gradient magnitude, reducing their influence relative to squared loss. Unlike pure absolute loss, the curve is differentiable at zero and at the two joining points.

中文对应答案

在−1到1之间画一段抛物线，两端平滑连接直线：$L(0)=0$，$L(\pm1)=1/2$，$L(\pm2)=3/2$。损失对残差的导数，在区间内为r，在区间外为$\mathrm{sign}(r)$。因此，极大输出残差产生的梯度绝对值有上界，相比平方损失，其影响受到限制。与纯绝对值损失不同，Huber曲线在0及两处连接点都可导。



按题干公式作图 / Plot of the given formula

原题差异 / Source note： 随卷答案图高度为题干公式的两倍；本图按公式，L(±1)=0.5。 / The supplied plot is scaled by two; this plot follows the formula.

MT013 · 残差L1+权重L2，各自在做什么?

> **题型：** 简答题 / Written response · 解释

考点： 回归、特征选择与残差损失

出处： 2020B期中Quiz Q13，原卷第6页（10分）

English question

Your friend has designed a new regression algorithm that minimizes the L1 norm of the error and L2 norm of the weights, given by the following optimization problem:

$$w^* = \arg\min_w \left\{ \sum_{i=1}^{N} |w^T x_i - y_i| + \lambda \|w\|_2^2 \right\}.$$

What properties would you expect this regression algorithm to have? Explain why.

中文题意

某回归方法同时最小化残差的L1范数与权重的L2平方：

$$w^* = \arg\min_w \left\{ \sum_{i=1}^{N} |w^T x_i - y_i| + \lambda \|w\|_2^2 \right\}.$$

预期它有什么性质？解释原因。

English answer

Absolute residual loss grows linearly, so large output errors have less influence than under squared loss. L2 regularization discourages large coefficients and can stabilize fitting. It normally shrinks weights rather than producing the exact sparsity associated with L1 weight penalties. The loss is convex but nonsmooth at zero residual; suitable regularization can improve generalization, without guaranteeing it.

中文对应答案

绝对残差损失线性增长，因此大输出误差的影响比平方损失下小。L2正则化抑制大系数，可以使拟合更稳定；它通常缩小权重，而不是像L1权重惩罚那样产生恰好为0的稀疏系数。该损失是凸的，但在残差为0处不光滑。合适的正则化可能改善泛化，但不能保证一定改善。

MT050 · 缺车比车多更糟：设计非对称损失

> **题型：** 简答题 / Written response · 设计、作图、解释

考点： 回归、特征选择与残差损失

出处： 2023A期中 Q12，原卷第9页（10分）；2025A期中 Q12，原卷第9页（10分）

English question

You are working for a bike–sharing company, and your job is to predict the number of bicycles that will be borrowed from a popular bike–sharing station each day. This prediction will be used to inform the logistics team about how many bikes should be sent to the station. The CEO tells you that it is okay to have too many bikes sent to the station, but it is not good if the station runs out of bikes, since the customers will use other bike–sharing companies.

You have a training dataset (x_i, y_i), i=1,…,N, of historical data that contains feature vectors x_i, and number of bikes borrowed y_i. Consider a linear regression function $f(x) = w^T x + b$, which is trained with the following optimization problem:

$$(\hat{w}, \hat{b}) = \arg\min_{w,b} \sum_{i=1}^{N} L(f(x_i), y_i).$$

Design a loss function that can help you to do the regression according to the requirements, and draw a plot. Explain how your loss achieves the desired requirements.

中文题意

你在一家共享单车公司工作，需要预测一个热门站点每天的借车量，供调度团队决定向该站点运送多少辆车。公司认为多送一些车可以接受，但缺车会让顾客转用其他公司的单车。

已有历史训练数据(x_i, y_i)、$i = 1, \ldots, N$，其中x_i是特征向量，y_i是借车量。线性回归函数$f(x) = w^T x + b$通过下式训练：

$$(\hat{w}, \hat{b}) = \arg\min_{w,b} \sum_{i=1}^{N} L(f(x_i), y_i).$$

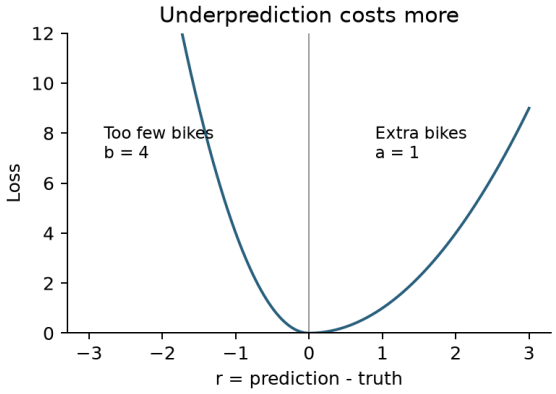
设计一个符合上述要求的损失函数，并画图。解释该损失怎样实现公司的要求。

English answer

Define $r = f(x) - y$. Use $L(r) = c_- r^2$ for $r < 0$ and $L(r) = c_+ r^2$ otherwise, with $c_- > c_+ > 0$. Draw two parabolic halves meeting at zero, the negative–residual side steeper. Equal–sized shortages then cost more than surpluses.

中文对应答案

定义残差$r = f(x) - y$。当$r < 0$时用$L(r) = c_- r^2$，否则用$L(r) = c_+ r^2$，其中$c_- > c_+ > 0$。画出两段在0处相接的抛物线，负残差一侧更陡。这样，同样数量的缺车会比车过剩受到更大的惩罚。



作图示例：图中 $b = c_- = 4$、$a = c_+ = 1$。 / Example plot with $b = c_- = 4$ and $a = c_+ = 1$.

MT037 · 线性回归为何与均值预测器表现相近?

> **题型：** 简答题 / Written response · 提出办法、解释

考点： 回归、特征选择与残差损失

出处： 2021B*期中 Q11，原卷第4页（10分）

English question

Your friend wants to predict the number of taxis used in HK in one hour from various features, such as weather, time of day, day of year, etc. Your friend uses linear regression to fit the data, but finds that the mean squared error (MSE) on the validation set is about the same as a “dummy” regressor that just predicts the mean of the training data. What two pieces of advice would you give to your friend to improve their regressor? Explain why.

中文题意

你的朋友想用天气、一天中的时段、一年中的日期等特征，预测香港一小时内的出租车使用量。他用线性回归拟合数据，却发现模型在验证集上的均方误差（MSE），与一个始终预测训练数据平均值的“基线”回归器相近。你会提出哪两个改进建议？分别说明原因。

English answer

Two useful directions are (1) add informative features or interactions, such as time–of–day effects interacting with weather, so the model can learn demand changes that the current features miss; and (2) use a nonlinear representation/model if the relationship is not linear, because an additive linear rule cannot represent curved or threshold–like demand patterns. Compare by validation or training–only CV. Also check excessive regularization, data alignment and optimization. The validation result alone does not prove the training least–squares solution is wrong.

中文对应答案

两个值得尝试的方向是：(1) 加入有用的特征或交互项，例如一天中的时段与天气之间的交互，让模型学到现有特征遗漏的需求变化；(2) 若输入与输出的关系不是线性的，就采用非线性表示或模型，因为简单线性相加无法表示弯曲或带阈值的需求变化。通过验证集或训练数据内部的交叉验证比较效果。同时检查正则化是否过强、数据是否正确对齐，以及优化是否正常。仅凭验证结果，不能认定训练得到的最小二乘解有错。

MT062 · 随机森林：树多了为什么仍有共同误差?

> **题型：** 简答题 / Written response · 读图或公式、提出办法

考点： 集成学习

出处： 2023B期中 Q11，原卷第8页（10分）

English question

For a random forest F of n trees, suppose the variance of the error of a decision tree f is σ^2, and the correlation coefficient between the errors of two trees is ρ. We will have the error variance of F as:

$$\mathrm{var}(F) = \mathrm{var}\left(\frac{1}{n}\sum_{i=1}^{n} f_i\right) = \frac{\sigma^2}{n} + \frac{n-1}{n}\rho\sigma^2.$$

(a) Explain the relationship between the number of trees and the correlation coefficient and the ability of F to reduce overfitting.

(b) For a fixed number of trees n, explain some concrete methods to reduce the error variance of F.

中文题意

随机森林F由n棵树组成。单棵决策树f的误差方差为σ^2，两棵不同树的误差相关系数为ρ。森林取各树预测的平均，其误差方差为

$$\mathrm{Var}(F) = \mathrm{Var}\left(\frac{1}{n}\sum_{i=1}^{n} f_i\right) = \frac{\sigma^2}{n} + \frac{n-1}{n}\rho\sigma^2.$$

(a) 解释树的数量、误差相关系数与森林减少过拟合能力的关系。

(b) 固定树的数量n时，说明几种降低森林误差方差的具体方法。

English answer

(a) Rewrite the variance as $\sigma^2[\rho + (1-\rho)/n]$. For fixed tree variance and fixed $0 \leq \rho \leq 1$, more trees reduce the second term; as $n \to \infty$, the variance approaches $\rho\sigma^2$. If $\rho = 0$, it is σ^2/n; if $\rho = 1$, adding trees gives no reduction. Lower correlation therefore makes averaging more effective at reducing the variance component of overfitting.

(b) At fixed n, train trees on different bootstrap samples and randomly choose feature candidates at each split. These changes reduce shared tree errors, while each tree should retain predictive strength. Lower variance alone does not guarantee lower total test error.

中文对应答案

(a) 将方差改写为$\sigma^2[\rho + (1-\rho)/n]$。固定单树方差和$0 \leq \rho \leq 1$时，增加树的数量会减小第二项；当$n \to \infty$时，方差趋于$\rho\sigma^2$。若$\rho = 0$，方差为$\sigma^2/n$；若$\rho = 1$，再加树也不能降低方差。因此，树之间的相关性越低，平均预测越能减轻与方差有关的过拟合。

(b) 固定n时，用不同的bootstrap样本训练各树，并在每次分裂时随机选择候选特征。这样可以减少各树共同犯的错误，同时应保留各棵树的预测能力。仅降低方差，不保证总测试误差一定降低。

MT038 · 100核部署：RF与AdaBoost如何并行预测?

> **题型：** 简答题 / Written response · 比较、提出办法

考点： 集成学习

出处： 2021B*期中 Q12，原卷第4页（10分）

English question

You are working for Tesla on their self-driving car software for their new model. Your goal is to classify objects from the radar scanner. After extensive testing, you find that AdaBoost and Random Forest classifiers both perform equally well at the task. They also both use equal amount of computation (flops). The Random Forest uses 10 trees with depth 10, while Adaboost has 100 weak learners. The new Tesla model has a special 100-core processor for running the self-driving car software. Which classifier do you recommend for implementation in the new car? Explain why.

中文题意

雷达目标分类中，RF与AdaBoost准确率相同、总计算量相同。RF有10棵深度10的树，AdaBoost有100个弱学习器；部署芯片有100个核心。推荐哪种？解释理由。

English answer

The intended choice is **AdaBoost**, assuming its 100 weak learners are shallow and can run independently at prediction time. They can occupy the 100 cores, whereas the 10 RF trees each retain a depth–10 serial path. Under an idealized one–comparison weak–learner model this gives lower latency. Both ensembles can parallelize inference; AdaBoost's sequential training does not forbid parallel prediction. Exact speed depends on learner complexity, scheduling and aggregation overhead.

中文对应答案

本题预期选择**AdaBoost**，前提是它的100个弱学习器足够浅，且预测时可以独立运行。它们可以分别占用100个核心；随机森林的10棵树则各自仍有一条深度为10、需要依次执行的路径。在每个弱学习器只做一次比较的理想化模型下，AdaBoost的延迟更低。两种集成都可以并行预测；AdaBoost训练需要依次进行，并不妨碍预测并行。实际速度仍取决于学习器复杂度、调度和结果汇总的开销。

MT049 · 在线更新AdaBoost：速度与异常值

> **题型：** 简答题 / Written response · 提出办法、判断并说明

考点： 集成学习

出处： 2023A期中 Q11，原卷第8页（10分）

English question

You are working on an online self–driving system that employs the Adaboost model for real–time decision making. The Adaboost model will be updated with newly collected data in the real–world while the system is running, which requires the model to converge fast. (a) How would you adjust the model or hyperparameters to make this real–time system? (b) Is using Adaboost a good choice in this situation? Why or why not?

中文题意

自动驾驶系统用AdaBoost实时决策，运行期间还要用新采集数据更新模型，因此希望快速收敛。(a) 怎样调整模型或超参数？(b) AdaBoost是否适合这种情境？解释。

English answer

(a) Limit the number and complexity of weak learners, and tune the learning rate to meet a validated latency/accuracy budget. A larger rate may need fewer rounds, but can destabilize learning or worsen noise sensitivity. (b) Conventional batch AdaBoost is not automatically an online–learning algorithm. Its exponential loss can emphasize difficult/noisy points; appending learners indefinitely also increases prediction cost. A bounded model, scheduled retraining or a method designed for streaming updates may be preferable.

中文对应答案

(a) 限制弱学习器的数量和复杂度，并调整学习率，通过验证找到满足延迟与准确率要求的设置。较大的学习率可能减少所需轮数，但也可能使学习不稳定，或对噪声更敏感。

(b) 常规批量AdaBoost并不会自动成为在线学习算法。它的指数损失可能过分强调难分或含噪声的点；无限追加学习器也会不断增加预测开销。限制模型规模、定期重训，或采用专门支持流式更新的方法，可能更合适。

MT064 · 综合题：Bagging/Boosting与Huber

> **题型：** 简答题 / Written response · 比较、作图、解释

考点： 集成学习

出处： 2025A期中 Q7，原卷第4页（10分）

English question

(a) [5 marks] Explain the association and difference between boosting and bagging.

(b) [5 marks] Consider linear regression using the Huber loss function:

$$L(f(x_i), y_i) = \begin{cases} \frac{1}{2}[y_i - f(x_i)]^2, & |y_i - f(x_i)| < 1, \\ |y_i - f(x_i)| - \frac{1}{2}, & \text{otherwise.} \end{cases}$$

Draw a plot of the loss function and answer whether the loss is robust to outliers with reason.

中文题意

(a) [5分] 解释boosting与bagging的联系和区别。

(b) [5分] 线性回归使用Huber损失：当 $|y_i - f(x_i)| < 1$ 时为 $\frac{1}{2}[y_i - f(x_i)]^2$，否则为 $|y_i - f(x_i)| - \frac{1}{2}$。画图，判断是否对异常值稳健并说明理由。

English answer

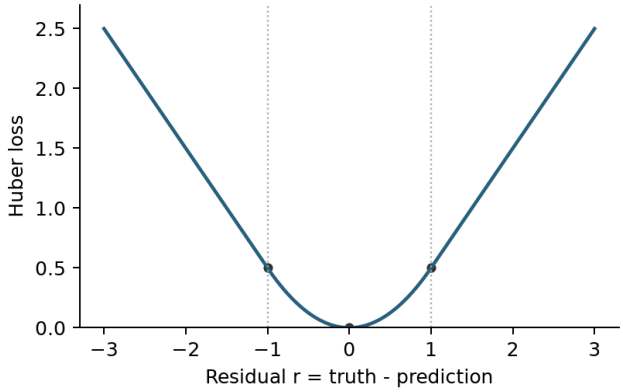
(a) Both combine learners. Bagging fits resampled learners independently; boosting trains sequentially using the current ensemble's errors or weighted loss.

(b) Let $r = y_i - f(x_i)$. Huber is quadratic inside $|r| < 1$ and linear outside. Join smoothly at $(\pm 1, 1/2)$, with tail slopes ± 1. Bounded residual gradients reduce sensitivity to large output errors.

中文对应答案

(a) 两者都组合多个学习器。Bagging在重采样数据上独立训练各个模型；boosting则根据当前集成的错误或加权损失，依次训练后续模型。

(b) 令残差$r = y_i - f(x_i)$。Huber损失在$|r| < 1$时是二次的，区间外是线性的。在$(\pm 1, 1/2)$处平滑连接，两侧直线的斜率分别为−1和+1。损失对残差的梯度有界，因此对大的输出误差不那么敏感。



按题干公式作图 / Plot of the given formula

MT008 · 平方形分类损失为什么惩罚“太正确”?

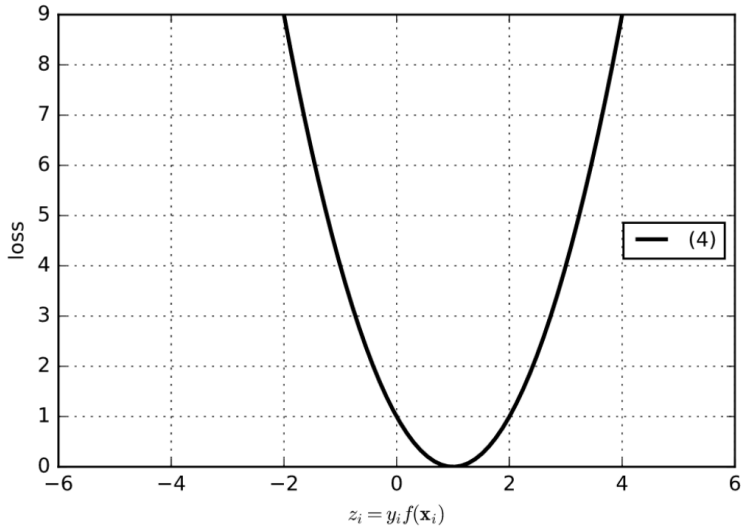
> **题型：** 简答题 / Written response · 读图或公式、判断并说明

考点： 分类损失图与错误代价

出处： 2020B期中Quiz Q8，原卷第4页（10分）

English question

Consider a classifier with the loss function in the below figure. Using this loss function, describe some expected properties of this classifier and explain why. Will this be a good classifier or a bad classifier? Why?



中文题意

观察题图中的分类损失。描述使用这种损失训练的分类器会有什么性质，并解释原因。你认为它是好的还是不好的分类损失？图中横轴为 $z_i = y_i f(x_i)$。

English answer

The squared-shaped loss targets z=1. Large tail gradients make it sensitive to extreme errors, while scores above 1 penalize confidently correct examples unnecessarily. Both effects can make fitting less robust, especially with noisy labels.

中文对应答案

这个平方形损失把$z = 1$作为目标。两端较大的梯度使它对极端错误敏感；分数超过1时，它又会不必要地惩罚高置信度的正确预测。这两点都可能使拟合不够稳健，尤其是在标签有噪声时。

MT026 · V形损失：最优点为何在z=1?

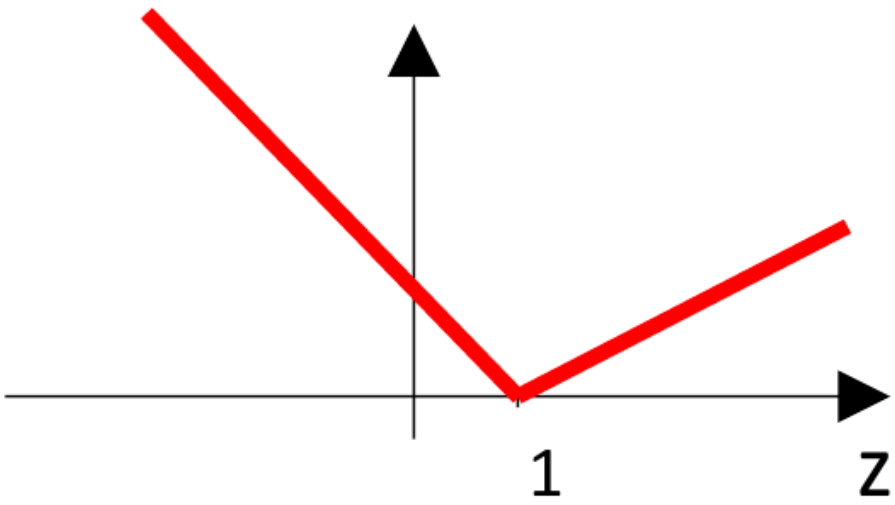
> **题型：** 简答题 / Written response · 读图或公式、判断并说明

考点： 分类损失图与错误代价

出处： 2021A期中 Q13，原卷第4页（10分）

English question

Consider the following classification loss function. Describe the properties of the classifier learned with this loss function. Will this be a good classifier? Why or why not? Note: z = y*f(x).



中文题意

读题图中的分类损失，描述所得分类器的性质。它是好的分类器吗？说明理由。横轴 $z = yf(x)$，图中两侧均为直线，在z=1处达到最小值。

English answer

The loss is minimized at z=1 and grows linearly on both sides. It corrects wrong or small–margin predictions with bounded tail slopes, but also penalizes confidently correct scores above 1. The latter is unnecessary for margin classification; it does not prove universally poor accuracy.

中文对应答案

损失在$z = 1$处最小，并向两侧线性增长。它能修正错误预测或间隔较小的预测，两端斜率的绝对值有界；但分数超过1时，也会惩罚高置信度的正确预测。后一种惩罚对基于间隔的分类没有必要，但这并不证明它在所有任务上准确率都很差。

MT051 · 负间隔也零损失：分类器会放过什么？

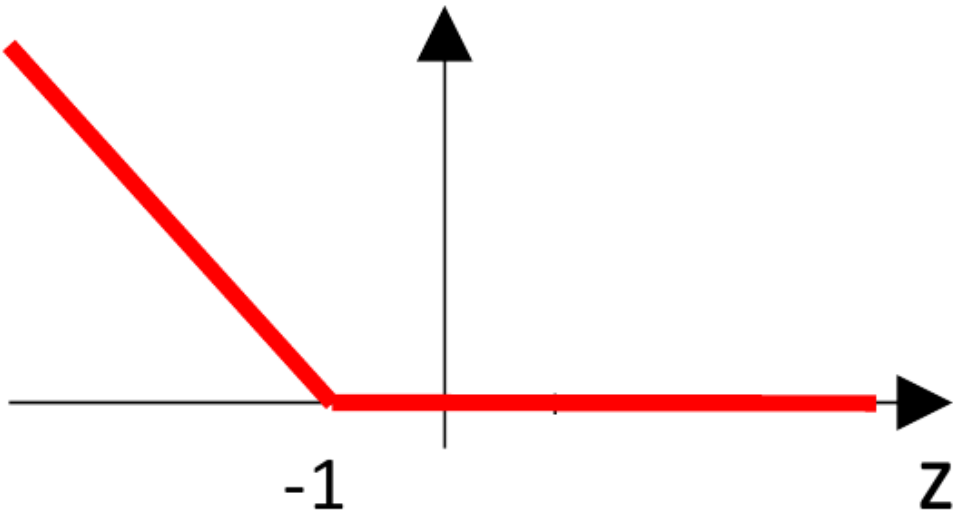
> **题型：** 简答题 / Written response · 读图或公式、判断并说明、作图

考点： 分类损失图与错误代价

出处： 2023A期中 Q13，原卷第10页（10分）

English question

Consider the following classification loss function. Describe the properties of the classifier learned with this loss function. Will this be a good classifier? Why or why not? Draw a figure of an example of a linear classifier trained with this loss. Note: z = y*f(x).



中文题意

描述题图所示分类损失训练出的分类器性质，判断它是否合适并说明理由。再画一幅采用此损失的线性分类器例子。横轴 $z = yf(x)$；曲线在z=−1处降至0，之后为0。

English answer

A representative function with this shape is $L(z) = \max(0, -1 - z)$; the vertical scale is not specified. Every −1⩽z<0 example is misclassified but has zero loss, so minimizing this objective need not correct it. It is therefore unsuitable as a sole objective for enforcing correct

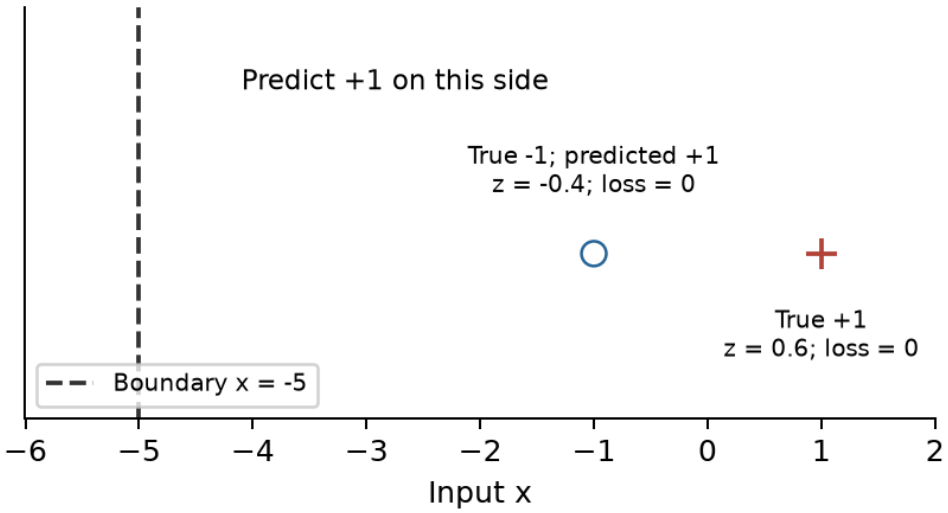
classification.

For an explicit example use f(x)=0.1x+0.5 with (x,y)=(−1,−1) and (1,+1). The signed scores are −0.4 and 0.6; both losses are zero, but the negative example is classified positive. Draw the boundary x=−5 and the two points to its right, with opposite true labels.

中文对应答案

具有这种形状的一个代表性函数是$L(z) = \max(0, -1 - z)$；原图没有指定纵轴尺度。所有满足$-1 \leq z < 0$的样本都被分错，却有零损失，因此最小化这个目标未必会纠正它们。它不适合作为促使模型正确分类的唯一目标。

具体例子可取$f(x) = 0.1x + 0.5$，两个样本为$(x, y) = (-1, -1)$和$(1, +1)$。带标签的分数分别为−0.4和0.6，损失都为0，但负类样本被判成了正类。画图时，把边界画在$x = -5$，两个真实标签相反的样本都位于边界右侧。



答案示例图 / Answer example

MT039 · 饱和的非凸分类损失

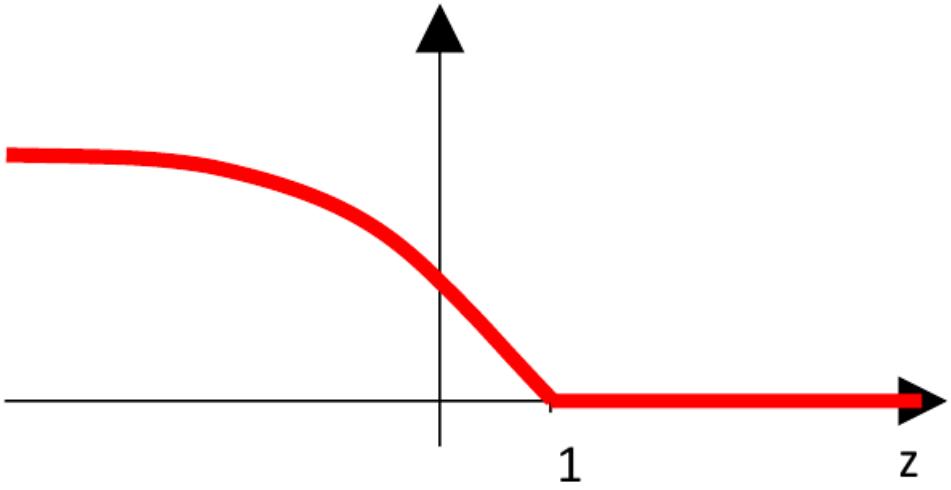
> **题型：** 简答题 / Written response · 读图或公式

考点： 分类损失图与错误代价

出处： 2021B*期中 Q13，原卷第4页（10分）

English question

Consider the following classification loss function. Describe the properties of the classifier learned with this loss function. Note: z = y*f(x).



中文题意

描述题图所示损失训练出来的分类器会有哪些性质。横轴 $z = yf(x)$；z≥1后损失为0，向很负的z方向，曲线逐渐变平。

English answer

The loss is zero for z≥1 and saturates at a high level for very negative z. Its left-tail slope approaches zero, limiting extreme points' gradient influence but potentially leaving wrong predictions unrecovered. The shape is nonconvex, making optimization harder. The decreasing quantity is slope magnitude, not the loss value.

中文对应答案

当$z \geq 1$时损失为0；当z是很大的负数时，损失在较高水平饱和。左端斜率趋于0，限制了极端样本对梯度的影响，但也可能使错误预测得不到纠正。这个形状是非凸的，优化更困难。减小的是斜率的绝对值，而不是损失值。

MT065 · 比较Logistic、Hinge和Exponential损失

> **题型：** 简答题 / Written response · 比较、排序

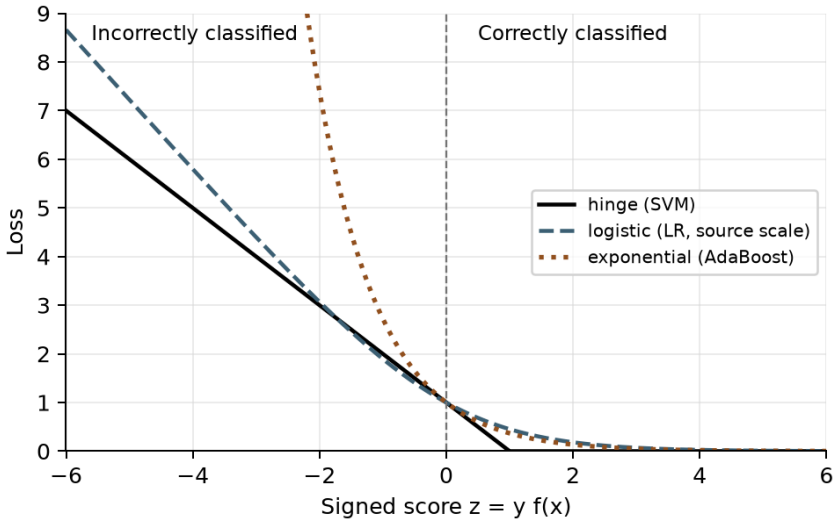
考点： 分类损失图与错误代价

出处： 2025A期中 Q9，原卷第6页（10分）

English question

Logistic Regression and Support Vector Machine (SVM), and Adaboost are all commonly used classifiers.

1. What is the key difference in the optimization objective of Logistic Regression and SVM? [6 marks]
2. Given this picture, please describe the three algorithms' sensitivity to outliers in descending order, and explain why. [4 marks]



按原卷曲线重绘并去掉手写批注；保留原图中LR的相对缩放，比较以题图为准。 / Clean redraw preserving the displayed relative scaling.

中文题意

Logistic Regression、SVM和AdaBoost都是常见分类器。

1. [6分] LR与SVM的优化目标主要有什么区别?
2. [4分] 根据题图，将三种算法对异常点的敏感程度从高到低排序，并解释。图中横轴为有符号分数z，负值表示错分，正值表示分对。

English answer

1. Logistic regression minimizes conditional log–loss, log(1+exp(−z)), usually with regularization. A soft–margin SVM minimizes a weight–norm penalty plus hinge loss max(0,1−z), balancing margin and violations.
2. **For the far–negative–margin region in the supplied figure: AdaBoost > LR > SVM.** Exponential loss has the steepest tail, so extreme misclassifications exert the strongest gradient influence. The plotted LR tail is steeper than the hinge tail. This comparison depends on the figure's loss scaling; it is not a universal ranking of the algorithms.

中文对应答案

1. 逻辑回归最小化条件对数损失$\log(1+\exp(-z))$，通常还带有正则化。软间隔SVM最小化权重范数惩罚与hinge损失$\max(0,1-z)$之和，在扩大间隔和允许违反约束之间作权衡。
2. **在原图的远端负间隔区域，影响大小为AdaBoost > LR > SVM。** 指数损失的曲线最陡，因此极端错分样本对梯度的影响最强。图中LR左端也比hinge损失更陡。这一比较取决于图中各损失的缩放方式，不是三种算法普遍成立的排名。

MT007 · 筛查任务：漏诊比误报更贵

> **题型：** 简答题 / Written response · 提出办法、解释

考点： 分类损失图与错误代价

出处： 2020B期中Quiz Q7，原卷第3页（10分）；2023B期中 Q13，原卷第10页（10分）

English question

You have been asked to build a classifier for an early screening test for diagnosing cancer. After the screening test, a more expensive advanced test will be used to confirm the result. For the screening test, you are given a feature vector, and the goal is to predict whether or not the person has cancer (positive or negative). Since it is a screening test, the classifier should classify all people with cancer as positive, while it is okay for some people without cancer to be misclassified as positive. Discuss two methods for getting this desired behavior from the classifier.

中文题意

设计癌症早期筛查二分类器。输入是特征向量，输出阳性或阴性；阳性者还会做更昂贵的确诊检查。目标是将所有癌症患者判为阳性，不漏掉病例，允许部分健康者被误报。讨论两种达到这种偏好的方法。

English answer

1. Give positive/cancer cases or false-negative errors a larger training cost, so missing a positive is penalized more.
2. Lower the threshold for predicting the positive class. If positive is predicted when f(x)>t, lower t; for a calibrated positive-class probability, lower the probability threshold below the usual 0.5 as appropriate.

Choose the threshold on validation data using sensitivity and false-positive cost. These choices favour recall but cannot guarantee zero false negatives on unseen patients.

中文对应答案

1. 在训练时，给阳性 / 癌症样本或假阴性错误更高的代价，使漏掉阳性受到更大的惩罚。
2. 降低预测为阳性的阈值。如果规则是$f(x) > t$时判为阳性，就降低t；如果输出是经过校准的阳性概率，则可适当把概率阈值降到通常的0.5以下。

应在验证数据上，结合灵敏度与假阳性的代价选择阈值。这些调整有利于提高召回率，但不能保证对未见过的患者零漏诊。

MT023 · 1000名受试者中只有50名阳性

> **题型：** 简答题 / Written response · 诊断问题、提出办法

考点： 分类损失图与错误代价

出处： 2021A期中 Q10，原卷第4页（10分）；2023A期中 Q10，原卷第7页（10分）

English question

You are working for a hospital to build a classifier as a cost–effective screening test for a lung disease based on the patient's demographic data and saliva sample. If the classifier predicts positive, then the patient will do a follow–up CT scan that is more costly and very accurate. You have collected a dataset of 1000 patients, of which 50 have the lung disease. What are the key issues to consider when training and testing your classifier? How do you address these key issues during training and testing?

中文题意

医院用人口信息和唾液样本做肺病筛查，阳性者再做更贵且准确的CT。数据共1000人，50人患病。训练和测试应考虑哪些关键问题？怎样处理？

English answer

There are two distinct issues: **class imbalance** (50 positives versus 950 negatives) and **asymmetric error costs** (missing disease is more costly than sending someone for CT). Use stratified training/validation splits and positive–class weighting or appropriate resampling within training. Select a lower positive threshold using validation data, and report sensitivity/recall, specificity and the confusion matrix rather than accuracy alone. Keep the test set separate.

中文对应答案

这里有两个不同的问题：**类别不平衡**，即50名阳性、950名阴性；以及**错误代价不对称**，即漏诊比让人多做一次CT的代价更高。训练集和验证集应分层划分，在训练数据内部使用阳性类别加权或合适的重采样。通过验证数据选择较低的阳性阈值，报告灵敏度 / 召回率、特异度和混淆矩阵，不能只报告准确率。测试集单独保留。

MT035 · 正常邮件被大量拦截怎么办?

> **题型：** 简答题 / Written response · 提出办法、解释

考点： 分类损失图与错误代价

出处： 2021B*期中 Q9，原卷第4页（10分）

English question

You have been asked to build a classifier for detecting junk email. You are given a training dataset, which has more junk email than regular email. You trained the model with logistic regression. However, your classification model predicts most emails as spam, so that you will

hardly receive any email at all. What are two approaches to fix this problem? Give reasons why these approaches will work.

中文题意

邮件分类训练集中垃圾邮件多于正常邮件。逻辑回归把大多数邮件都判成垃圾，几乎收不到邮件。提出两种修复方法并说明理由。

English answer

Increase the cost/weight of misclassifying regular email during training, so blocking a legitimate message incurs a larger penalty. Also raise the threshold for calling an email spam: if $p = P(\text{spam} \mid x)$ and spam is predicted when $p > t$, increasing t requires stronger evidence before blocking a message. Collecting more representative regular–mail training data is another option. Select the tradeoff on validation data and track false positives on regular mail.

中文对应答案

训练时提高把正常邮件误判为垃圾邮件的代价或权重，让误拦正常邮件受到更大的惩罚。同时提高判为垃圾邮件的阈值：若$p = P(\text{spam} \mid x)$，且$p > t$时判为垃圾邮件，那么提高$t$就要求模型有更强的证据才拦截邮件。也可以增加有代表性的正常邮件训练数据。通过验证数据选择取舍，并持续查看正常邮件被误拦截的情况。

MT067 · 链式求导与两层网络反传

> **题型：** 简答题 / Written response · 计算推导

考点： 神经网络与链式法则

出处： 2025A期中 Q13，原卷第10页（10分）

English question

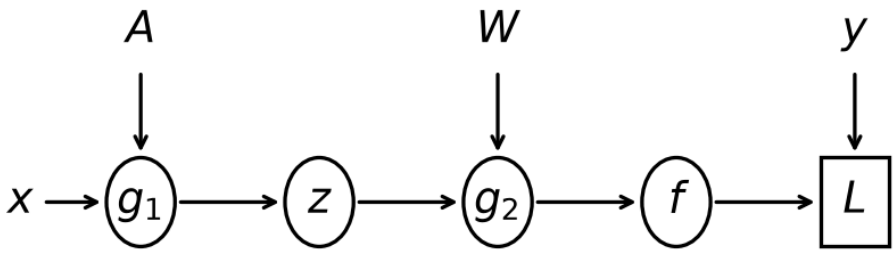
(a) Warm–up question (you need to give detailed steps; you may define intermediate variables accordingly): calculate the derivative of y with respect to x using the chain rule.

$$y = \left(1 + e^{\sqrt[3]{3x^2} + \tan(5x)}\right)^{-1}.$$

(b) Please first explain the forward process with respect to the graph below and then calculate

$$\frac{\partial L}{\partial a_i}, \qquad \frac{\partial L}{\partial g_1}, \qquad \frac{\partial L}{\partial w_j}.$$

Hint: $A \in \mathbb{R}^{m \times m}$ and $W \in \mathbb{R}^{m \times n}$ are matrices; a_i and w_j are general representations of each weight vector in A and W, respectively. For $g_1 \to z$ and $g_2 \to f$, any proper functions are fine.



按原卷重绘的前向结构示意图；省略原图中的反向梯度提示，保留本题所需节点与参数箭头。 / Simplified forward graph redrawn from the exam; backward-gradient annotations are omitted, while the required nodes and parameter arrows are retained.

中文题意

(a) 用链式法则求下式对x的导数，给出详细步骤，可定义中间变量：

$$y = \left(1 + e^{\sqrt[3]{3x^2} + \tan(5x)}\right)^{-1}.$$

(b) 解释题图的前向过程，然后求 $\partial L/\partial a_i$、$\partial L/\partial g_1$ 和 $\partial L/\partial w_j$。题设 $A \in \mathbb{R}^{m\times m}$、$W \in \mathbb{R}^{m\times n}$；$a_i, w_j$分别表示相应矩阵中的权重向量。允许为 $g_1 \to z$、$g_2 \to f$ 选择合适函数。

English answer

(a) Let $v = 3x^2$, $u = v^{1/3} + \tan(5x)$ and $y = (1 + e^u)^{-1}$. Where $x \neq 0$ and $\cos(5x) \neq 0$, the inner derivatives are

$$\frac{dv}{dx} = 6x, \qquad \frac{d(v^{1/3})}{dv} = \frac{1}{3}v^{-2/3}, \qquad \frac{d\tan(5x)}{dx} = 5\sec^2(5x).$$

Apply the chain rule to each part of u, then to y:

$$\frac{du}{dx} = \frac{1}{3}v^{-2/3}(6x) + 5\sec^2(5x) = \frac{2x}{(3x^2)^{2/3}} + 5\sec^2(5x),$$

$$\frac{dy}{du} = -(1 + e^u)^{-2}e^u, \qquad \frac{dy}{dx} = -\frac{e^u}{(1 + e^u)^2}\left[\frac{2x}{(3x^2)^{2/3}} + 5\sec^2(5x)\right].$$

(b) Choose a tanh hidden layer and a linear output, as permitted, and leave the unspecified loss as $L(f, y)$. Treat a_i and w_j as columns:

$$g_1 = A^T x, \quad z = \tanh(g_1), \quad g_2 = W^T z, \quad f = g_2, \quad L = L(f, y).$$

Let $\delta_2 = \nabla_f L$. Since $f_j = w_j^T z$ and $dz_i/dg_{1i} = 1 - z_i^2$, the chain rule gives

$$\frac{\partial L}{\partial z_i} = \sum_{j=1}^{n} W_{ij}\delta_{2j}, \qquad \delta_{1i} = \frac{\partial L}{\partial g_{1i}} = (1 - z_i^2)\sum_{j=1}^{n} W_{ij}\delta_{2j}.$$

Thus $\delta_1 = (W\delta_2) \odot (1 - z^2)$, where $\odot$ denotes elementwise multiplication. Because $g_{1i} = a_i^T x$ has derivative x with respect to a_i, and $f_j = w_j^T z$ has derivative z with respect to w_j,

$$\frac{\partial L}{\partial g_1} = \delta_1, \qquad \frac{\partial L}{\partial a_i} = x\,\delta_{1i}, \qquad \frac{\partial L}{\partial w_j} = z\,\delta_{2j}.$$

Here δ_1 and x are $m-$dimensional, δ_2 is $n-$dimensional and each column gradient is $m-$dimensional. For half–squared error, $\delta_2 = f - y$; another loss requires its own derivative.

中文对应答案

(a) 令$v = 3x^2$，$u = v^{1/3} + \tan(5x)$，$y = (1 + e^u)^{-1}$。在$x \neq 0$且$\cos(5x) \neq 0$处，内层各项的导数为

$$\frac{dv}{dx} = 6x, \qquad \frac{d(v^{1/3})}{dv} = \frac{1}{3}v^{-2/3}, \qquad \frac{d\tan(5x)}{dx} = 5\sec^2(5x).$$

先对u的各部分使用链式法则，再对y求导：

$$\frac{du}{dx} = \frac{1}{3}v^{-2/3}(6x) + 5\sec^2(5x) = \frac{2x}{(3x^2)^{2/3}} + 5\sec^2(5x),$$

$$\frac{dy}{du} = -(1 + e^u)^{-2}e^u, \qquad \frac{dy}{dx} = -\frac{e^u}{(1 + e^u)^2}\left[\frac{2x}{(3x^2)^{2/3}} + 5\sec^2(5x)\right].$$

(b) 按题目允许的选择，隐藏层使用tanh，输出层使用线性函数；题目未指定的损失保留为$L(f, y)$。把a_i和w_j视为列向量：

$$g_1 = A^T x, \quad z = \tanh(g_1), \quad g_2 = W^T z, \quad f = g_2, \quad L = L(f, y).$$

令$\delta_2 = \nabla_f L$。由$f_j = w_j^T z$及$dz_i/dg_{1i} = 1 - z_i^2$，链式法则给出

$$\frac{\partial L}{\partial z_i} = \sum_{j=1}^{n} W_{ij}\delta_{2j}, \qquad \delta_{1i} = \frac{\partial L}{\partial g_{1i}} = (1 - z_i^2)\sum_{j=1}^{n} W_{ij}\delta_{2j}.$$

因此$\delta_1 = (W\delta_2) \odot (1 - z^2)$，其中⊙表示逐元素相乘。$g_{1i} = a_i^T x$对$a_i$的导数是$x$，$f_j = w_j^T z$对$w_j$的导数是$z$，所以

$$\frac{\partial L}{\partial g_1} = \delta_1, \qquad \frac{\partial L}{\partial a_i} = x\,\delta_{1i}, \qquad \frac{\partial L}{\partial w_j} = z\,\delta_{2j}.$$

其中δ_1和x是m维向量，δ_2是n维向量，每个列参数的梯度都是m维。若损失为平方误差的一半，则$\delta_2 = f - y$；若采用其他损失，需要使用相应的导数。

MT066 · 梯度消失：为什么深度和激活函数有关?

> **题型：** 简答题 / Written response · 解释、计算推导

考点： 神经网络与链式法则

出处： 2025A期中 Q10，原卷第7页（10分）

English question

(a) [4 marks] Explain what is the “vanishing gradient” problem.

(b) [3 marks] Explain mathematically why “deeper” networks are more prone to this problem. You may use equations or a mathematical description.

(c) [3 marks] Compared with the Sigmoid activation function, why can the ReLU activation function alleviate this problem?

中文题意

(a) [4分] 解释梯度消失问题。(b) [3分] 用数学说明为什么更深网络更容易出现这个问题。(c) [3分] 相比Sigmoid，ReLU为什么能缓解它?

English answer

(a) Gradients reaching earlier layers can become very small, so those parameters learn slowly. (b) Backpropagation multiplies local Jacobians. On a scalar path, $\partial h_L/\partial h_0$ is the product of weight and activation derivatives. Sigmoid derivatives are at most 1/4; with unit weights, a path through L sigmoids has gradient magnitude at most $(1/4)^L$. (c) An active ReLU has derivative 1, avoiding sigmoid's saturating factor on that path. Negative ReLU inputs have derivative zero, and weights still affect the product, so ReLU is not a universal cure.

中文对应答案

(a) 传到前面各层的梯度可能变得很小，使这些层的参数学习得很慢。

(b) 反向传播会连乘各处的局部Jacobian矩阵。在一条标量路径上，$\partial h_L/\partial h_0$是沿途权重与激活函数导数的乘积。Sigmoid的导数最大为1/4；若权重均为1，经过L个sigmoid的路径，其梯度绝对值至多为$(1/4)^L$。

(c) ReLU处于激活状态时导数为1，在这条路径上避免了sigmoid带来的饱和衰减因子。但ReLU输入为负时导数为0，权重也仍会影响连乘结果，因此ReLU不能普遍解决所有梯度消失问题。

MT036 · 历史拓展：音乐地理回归，GP还是RF?

> **题型：** 简答题 / Written response · 比较

考点： 历史范围拓展：Gaussian Process

出处： 2021B*期中 Q10，原卷第4页（10分）

历史期中原题；GPR属于本轮讲义主线之外的背景。

English question

Geographical origins of music are related to the audio features in music. Suppose we have 10,597 music segments from different regions of the world, where each audio segment is represented by a 68–dimensional vector, and the region is denoted by latitude and longitude coordinates. We know there are some outliers in the dataset and some regions have apparently more music data than others. Suppose we want to choose Gaussian process regression or random forest regression to predict the region coordinates from the audio feature vector. For this specific regression task, what are the advantages and disadvantages of using these Gaussian process regression or random forest regression?

中文题意

共有10,597段世界各地音乐，每段68维音频特征，目标是经纬度。存在异常值，且部分地区样本远多于其他地区。比较Gaussian Process Regression与Random Forest Regression在这个任务中的优缺点。

English answer

A standard GP provides a predictive mean and model–based uncertainty, useful when coverage is uneven; exact dense fitting on 10,597 samples is computationally costly, and a Gaussian likelihood can be sensitive to output outliers. RF captures nonlinear interactions and is often easier to scale, but ordinary leaf averages can still be affected by outlier targets and do not automatically provide calibrated uncertainty. Evaluate both with region–aware validation; sparse regions and out–of–range predictions remain difficult.

中文对应答案

标准GP提供预测均值及基于模型的不确定性，在数据覆盖不均匀时很有用。但对10,597个样本进行精确稠密拟合，计算成本很高；高斯似然也可能对输出异常值敏感。随机森林能捕捉非线性交互，通常更容易扩展到大数据，不过普通的叶节点平均值仍会受异常目标值影响，也不会自动给出经过校准的不确定性。应在验证时考虑地域划分，比较两者；样本稀少的地区和训练范围之外的预测，对它们仍然困难。

原卷索引 / Original-paper index

仅列本册题型在各原卷中的位置；未列题目在另一题型分册。按原卷题号查找本册位置；不同年份的重复题指向同一MT编号。2021B*保留题纸与配套答案的年份冲突。

2020B期中Quiz

原题	本册题号与页码	分值
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Q8	MT008（本册第29页）	10
Q9	MT009（本册第18页）	10
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Q12	MT012（本册第13页）	10
Q13	MT013（本册第22页）	10

2021A期中

原题	本册题号与页码	分值
Q7	MT020（本册第3页）	10
Q8	MT021（本册第9页）	10
Q9	MT022（本册第4页）	10
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2021B*期中

原题	本册题号与页码	分值
Q7	MT033（本册第7页）	10
Q8	MT034（本册第6页）	10
Q9	MT035（本册第36页）	10
Q10	MT036（本册第40页）	10
Q11	MT037（本册第24页）	10
Q12	MT038（本册第26页）	10
Q13	MT039（本册第32页）	10

2023A期中

原题	本册题号与页码	分值
Q7	MT046（本册第5页）	10
Q8	MT047（本册第11页）	10
Q9	MT048（本册第14页）	10
Q10	MT023（本册第36页）	10
Q11	MT049（本册第27页）	10
Q12	MT050（本册第23页）	10
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2023B期中

原题	本册题号与页码	分值
Q7	MT058（本册第10页）	10
Q8	MT059（本册第8页）	10
Q9	MT060（本册第15页）	10
Q10	MT061（本册第19页）	10
Q11	MT062（本册第25页）	10
Q12	MT063（本册第21页）	10
Q13	MT007（本册第35页）	10

2025A期中

原题	本册题号与页码	分值
Q7	MT064（本册第28页）	10
Q8	MT047（本册第11页）	10
Q9	MT065（本册第33页）	10
Q10	MT066（本册第39页）	10
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Q13	MT067（本册第37页）	10